1.2 Measurements & Errors

Question Paper

Course	CIE A Level Physics
Section	1. Physical Quantities & Units
Topic	1.2 Measurements & Errors
Difficulty	Medium

Time allowed: 80

Score: /65

Percentage: /100

Question 1a

The speed v of a transverse wave on a uniform string is given by the expression

$$v = \sqrt{\frac{Tl}{m}}$$

where T is the tension in the string, l is its length and m is its mass.

An experiment is performed to determine the speed v of the wave. The measurements are shown in Table 1.1.

Table 1.1

quantity	measurement	uncertainty
T	1.8 N	± 5%
l	126 cm	± 1%
m	5.1 g	± 2%

(a)
Use the data in Table 1.1 to calculate the speed v.

v = ... m s^{-1}

[2 marks]

Question 1b

Use your answer in **(a)** and the data in Table 1.1 to calculate the absolute uncertainty in the value of v.

absolute uncertainty = ... m s^{-1}

[2 marks]

Question 2a

A student used the apparatus shown to investigate the time taken for a hollow cylinder to roll down a ramp as shown in Fig. 1.1.

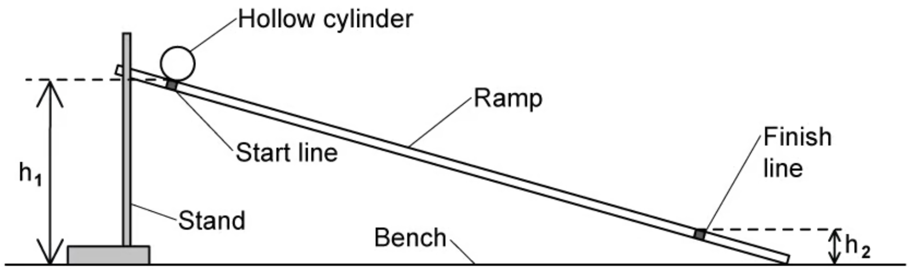


Fig. 1.1

(a)
State the most appropriate instrument for the measurement of the following, and describe how they can ensure the measurements are as accurate as possible

(i)
the heights h_1 and h_2

[2]

(ii)
the time taken for the cylinder to roll down the ramp

[2]

[4 marks]

Question 2b

The student wants to find the difference in height Δh between the start line and finish line

The heights are recorded as

$$h_1 = 67.5 \pm 0.1 \text{ cm}$$

$$h_2 = 8.6 \pm 0.1 \text{ cm}$$

(b)
Determine the change in height Δh, quoting your answer with its uncertainty

Δh = ………………… ± ………………… cm

[2 marks]

Question 2c

The student placed the cylinder on the start line and released it. She measured the time t for the cylinder to roll to the finish line. She repeated the measurements several times as shown in Table 1.1.

Table 1.1

t / s	0.88	0.82	0.80	0.86

(i)
Use the data in Table 1.1 to calculate the mean value of t and its uncertainty.

mean value of t = ... ± .... [2]

(ii)
Comment on the accuracy and precision of the data the student collected.

[2]

(iii)
Explain how reducing the value of h_1 could improve the measurement of t.

[2]

[6 marks]

Question 2d

The relationship between *t* and the acceleration of free fall *g* is given by

$$t^2 = \frac{4s^2}{g\Delta h}$$

where *s* is the distance along the ramp between the start line and the finish line.

The student recorded the ramp length as s = 102.0 cm ± 0.1 cm

(i)
Determine a value for *g*.

[2]

(ii)
Determine the percentage uncertainty in the value of *g*.

[2]

(iii)
Deduce whether the value of *g* is accurate.

[2]

[6 marks]

Question 3a

A cube made from aluminium is placed on a horizontal surface, as shown in Fig. 1.1.

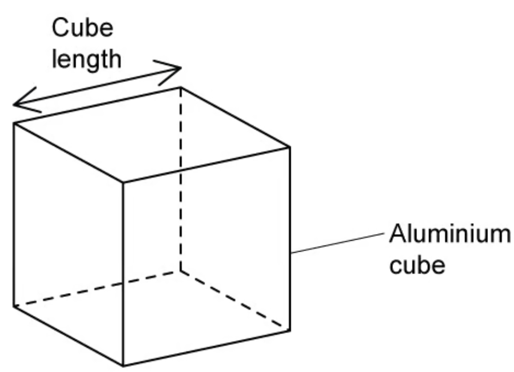


Fig. 1.1

The following measurements were made on the cube:

mass = 0.59 ± 0.01 kg

length of one side = 6.0 ± 0.1 cm

(a) Calculate the pressure produced by the cube on the surface.

[3 marks]

Question 3b

(b)

(i)

Calculate the percentage uncertainty in the pressure.

[2]

(ii)

State the pressure, with its actual uncertainty.

pressure = ... ± .... Pa [2]

[4 marks]

Question 3c

Calculate the density of aluminium with its uncertainty.

Express your answer to an appropriate number of significant figures.

density .... ± .... g cm^{-3}

[4 marks]

Question 3d

The student uses vernier callipers to measure the length of the side of the cube. After completing the experiment, he realises that when the callipers are fully closed, the reading is not zero.

(i)
State and explain whether this introduces a random error or a systematic error in the readings of the length.

[2]

(ii)
Explain why the readings are precise but not accurate.

[2]

[4 marks]

Question 4a

Analogue voltmeters and cathode-ray oscilloscopes (c.r.o.) are both devices which can measure potential differences.

(a)
For the analogue voltmeter and the c.r.o., describe **one** example of:

(i)
a systematic error that can affect both devices

[1]

(ii)
a systematic error that can only affect the analogue voltmeter

[1]

(iii)
a random error that can affect both devices

[1]

(iv)
a random error that can only affect the analogue voltmeter.

[1]

[4 marks]

Question 4b

The potential difference across a resistor is measured using the analogue voltmeter shown in Fig. 1.1.

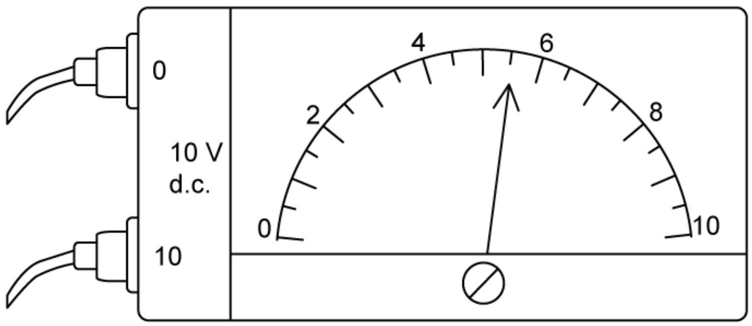


Fig. 1.1

(b)

The resistor is labelled as having a resistance of 125 Ω ± 3%.

(i)

Calculate the power dissipated by the resistor.

[2]

(ii)

Calculate the percentage uncertainty in the calculated power.

[2]

(iii)

Determine the value of the power, with its absolute uncertainty, to an appropriate number of significant figures.

power = ... ± .... W [2]

[6 marks]

Question 4c

A microphone connected to a cathode ray oscilloscope (c.r.o.) detects a sound wave of frequency *f*. The signal from the microphone is observed on the c.r.o. as shown in Fig. 1.2.

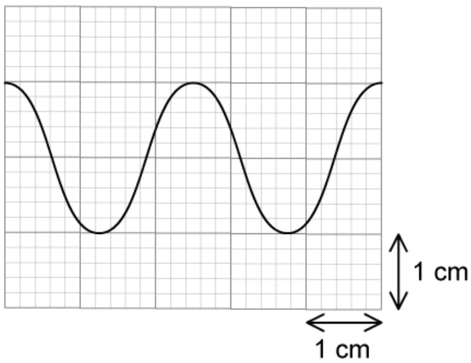


Fig. 1.2

The time-base setting of the c.r.o. is 0.50 ms cm^{-1}. The Y-plate setting is 2.5 mV cm^{-1}.

(c)
Using the trace in Fig. 1.2, determine

(i)
the amplitude of the signal, in mV

amplitude = ... mV [2]

(ii)
the frequency, in Hz

frequency = ... Hz [3]

[5 marks]

Question 4d

(d)

(i)

Determine the percentage uncertainty in amplitude caused by reading the scale on the c.r.o.

[2]

(ii)

State the amplitude with its actual uncertainty.

amplitude = ... ± .... mV [1]

[3 marks]

Question 5a

A student participates in an experiment to measure the Earth's gravitational field strength *g*. This is done using a simple pendulum.

The student suggests the period of oscillation *T* is related to the length of the pendulum *L* by the equation:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Table 1.1 shows the ten repeat readings of period *T*.

Table 1.1

0.67	0.66	0.67	0.68	0.69	0.64	0.66	0.65	0.68	0.65

Determine the mean period of oscillation and its percentage uncertainty.

[3 marks]

Question 5b

In a new experiment, the length of the pendulum L is measured with an accuracy of 1.8% and the acceleration due to free-fall g is measured with an accuracy of 1.6%.

(b)
If the time for the pendulum to complete 20 oscillations is 18.4 s, determine the time period for one oscillation and the absolute uncertainty in this value.

[3 marks]

Question 5c

Measurements of time periods for different lengths of pendula were taken using a stopwatch and plotted on the graph shown in Fig. 1.1.

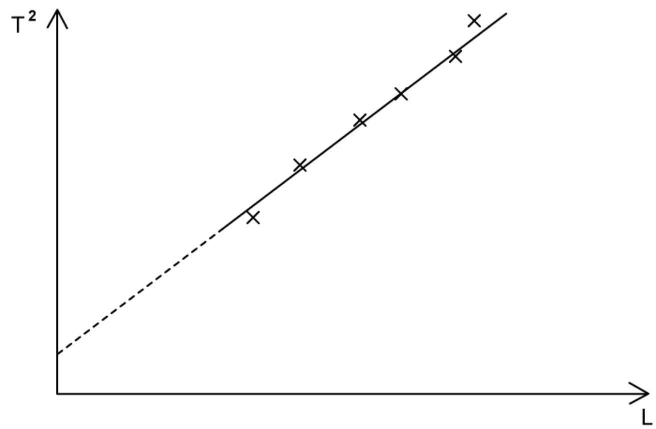


Fig. 1.1

(c)
Explain how the graph in Fig. 1.1 indicates that the readings are subject to systematic and random uncertainties.

[2 marks]

Question 5d

The period *T* for a mass *m* hanging on a spring performing simple harmonic motion is given by the equation:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Such a system is used to determine the spring constant *k*. The fractional error in the measurement of the period *T* is α and the fractional error in the measurement of the mass *m* is β.

(d)
Determine the fractional error in the calculated value of *k* in terms of α and β.

[2 marks]

Model Answers: Medium

1a

(a) Use the data in Table 1.1 to calculate the speed v:

List the known quantities and convert into SI base units:

- T = 1.8 N
- l = 126 cm = 126 × 10^{-2} m
- m = 5.1 g = 5.1 × 10^{-3} kg

Substitute the known quantities into the equation given:

- $v = \sqrt{\frac{Tl}{m}}$
- $v = \sqrt{\frac{1.8 \times (126 \times 10^{-2})}{(5.1 \times 10^{-3})}}$ [1 mark]
- v = 21.1 m s^{-1} = 21 m s^{-1} (2 s.f.) [1 mark]

[Total: 2 marks]

1b

(b) Calculate the absolute uncertainty in the value of v:

List the known quantities:

- Uncertainty in T = ± 5%
- Uncertainty in l = ± 1%
- Uncertainty in m = ± 2%

Calculate the percentage uncertainty in v:

- From the equation $v = \sqrt{\frac{Tl}{m}} = \left(\frac{Tl}{m}\right)^{\frac{1}{2}}$

- % uncertainty in v = ½ × (5% + 1% + 2%) = 4 % [1 mark]

Calculate the absolute uncertainty in v:

- Absolute uncertainty in $v = 4\% \times v = 0.04 \times 21$
- Absolute uncertainty in $v = 0.84 \text{ m s}^{-1}$ [1 mark]

[Total: 2 marks]

- T and l are multiplied together and divided by m
  - So, their percentage uncertainties are added together
  - 5% + 1% + 2% = 8%

- The result is then put to the power of ½
  - So the percentage uncertainty is halved
  - 8% × ½ = 4%

- Learn the **rules** for **combining uncertainties** from the **revision notes**

2a

(a)

(i) To measure heights h_1 and h_2:

- Use a metre ruler; [1 mark]

To ensure the measurements are as accurate as possible are:

Any **one** from:

- Place the rule as close as possible to the ramp; [1 mark]
- Use a set square / spirit level to ensure the rule is vertical; [1 mark]
- Ensure the rule reads zero at the bench; [1 mark]
- Avoid parallax error e.g. use a set square / read the scale at eye level; [1 mark]

(ii) To measure the time taken:

- Use a timer / stopwatch; [1 mark]

To ensure the measurements are as accurate as possible are:

Any **one** from:

- Start the timer immediately once the cylinder is released; [1 mark]
- Use a (fiducial) marker to mark the endpoint (to stop the timer at the same time); [1 mark]
- Repeat the experiment several times and take an average; [1 mark]

[Total: 4 marks]

2b

(b) The change in height Δ*h* is:

- $\Delta h = h_1 - h_2$ = 67.5 – 8.6 = 58.9 cm

The uncertainty in Δ*h* is:

- The uncertainty in each measurement is 0.1 cm
- Since the values of *h* are subtracted, the uncertainty is 0.1 cm + 0.1 cm = 0.2 cm [1 mark]

Quote Δ*h* with its uncertainty:

- Δ*h* = 58.9 ± 0.2 cm [1 mark]

[Total: 2 marks]

2c

(c)

(i) Calculate the mean value of *t* and its uncertainty

- Mean value of *t*: $\frac{0.88 + 0.82 + 0.80 + 0.86}{4} = 0.84$ s [1 mark]
- Uncertainty in *t*: $\pm\frac{1}{2}(range) = \pm\frac{1}{2}(0.88 - 0.80) = \pm 0.04$ s [1 mark]

(ii) Comment on the accuracy and precision of the data:

- Precision describes how close a set of measurements are to one another
- The readings are precise as they are reasonably close together; [1 mark]
- Accuracy describes how close the readings are to the true value
- The level of accuracy of the data cannot be determined (yet) **OR** it cannot be told whether the readings are close to the true value (yet); [1 mark]

(iii) Reducing h_1 might improve the measurement of *t* because:

- The values of *t* will increase **OR** the cylinder will move more slowly; [1 mark]

As well as any **one** from:

- So the percentage uncertainty in *t* will reduce; [1 mark]
- It will be easier to judge when the cylinder crosses the finish line; [1 mark]
- The effect of reaction time will be reduced; [1 mark]

[Total: 6 marks]

2d

(d)

List the known quantities:

- Change in height, $\Delta h = 58.9 \pm 0.2$ cm
- Time taken, $t = 0.84 \pm 0.04$ s
- Ramp length, $s = 102.0 \pm 0.1$ cm

(i) Determine a value for *g:*

- $t^2 = \dfrac{4s^2}{g\Delta h} \quad \Rightarrow \quad g = \dfrac{4s^2}{t^2 \Delta h}$
- $g = \dfrac{4 \times (102 \times 10^{-2})^2}{(0.84)^2 \times (58.9 \times 10^{-2})}$ [1 mark]
- $g = 10.014 = 10.0$ m s^{-2} [1 mark]

(ii) Determine the percentage uncertainty in the value of *g*

- When multiplying data, add the fractional or percentage uncertainties

(ii) Determine the percentage uncertainty in the value of *g*

- When multiplying data, add the fractional or percentage uncertainties
- When a quantity is raised to a power, multiply the uncertainty by the power
- Fractional uncertainty in *g*: $\frac{\Delta g}{g} = \left(2 \times \frac{\Delta s}{s}\right) + \frac{\Delta(\Delta h)}{\Delta h} + \left(2 \times \frac{\Delta t}{t}\right)$
- % uncertainty in *g* = $\left[2\left(\frac{0.1}{102}\right) + \left(\frac{0.2}{58.9}\right) + 2\left(\frac{0.04}{0.84}\right)\right] \times 100\%$ [1 mark]
- % uncertainty in *g* = 10.059 = 10 % [1 mark]

(iii) Deduce whether the measured value of *g* is accurate

Calculate the lower limit of *g*:

- Lower limit of g: *g* × (1 – 0.10) = 10 × 0.90 = 9.0 m s^{-2} [1 mark]

Write a conclusion:

- The measured value of *g* is 10.0 ± 1.0 m s^{-2} and the accepted value of *g* is 9.81 m s^{-2} hence, the value is accurate (since this lies within the lower limit of *g*); [1 mark]

[Total: 6 marks]

When you see command words such as 'deduce', make sure to write a **conclusion** at the end in order to score top marks

Knowing how to carry out the calculation is important, but examiners also want to see if you understand what you have calculated!

3a

(a) Calculate the pressure produced by the cube on the surface:

List the known quantities:

- Mass, *m* = 0.59 kg
- Length of cube, *L* = 6.0 cm = 0.06 m

Determine the pressure from the force and the area in contact with surface:

- Force on surface, F = weight = mg = 0.59 × 9.81
  AND
- Area in contact with surface, $A = L^2 = (0.06)^2$ [1 mark]
- Pressure: $P = \frac{F}{A}$
- $P = \frac{mg}{L^2} = \frac{0.59 \times 9.81}{(0.06)^2}$ [1 mark]
- P = 1607.75 = 1610 Pa (3 s.f.) [1 mark]

[Total: 3 marks]

3b

(b)

List the known quantities:

- Mass, m = 0.59 ± 0.01 kg
- Cube length, L = 6.0 ± 0.1 cm
- Pressure, P = 1600 Pa (from **(a)**)

(i) Calculate the percentage uncertainty in the pressure:

- When multiplying data, add the fractional or percentage uncertainties
- When a quantity is raised to a power, multiply the uncertainty by the power

- Pressure: $P = \frac{mg}{L^2}$
- Fractional uncertainty in P: $\frac{\Delta P}{P} = \frac{\Delta m}{m} + \left(2 \times \frac{\Delta L}{L}\right)$
- % uncertainty in $P = \left[\left(\frac{0.01}{0.59}\right) + 2\left(\frac{0.1}{6.0}\right)\right] \times 100\%$ [1 mark]
- % uncertainty in P = 5.028 = 5 % [1 mark]

(ii) State the pressure, with its actual uncertainty:

- Actual uncertainty in P = 1600 × 0.05 = 80 Pa [1 mark]
- Pressure = 1600 ± 80 Pa [1 mark]

Other acceptable answers for pressure = 1610 ± 80 or 1608 ± 81 Pa

[Total: 4 marks]

3c

(c) Calculate the density of aluminium, in g cm^{-3}, with its uncertainty:

List the known quantities:

- Mass, m = 0.59 ± 0.01 kg = 590 ± 10 g
- Cube length, L = 6.0 ± 0.1 cm

Calculate the density in g cm^{-3}:

- Density: $\rho = \frac{m}{V}$
- $\rho = \frac{m}{L^3} = \frac{590}{(6.0)^3}$
- ρ = 2.731 = 2.7 g cm^{-3} [1 mark]

Determine the uncertainty in density:

- Uncertainty in ρ: $\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \left(3 \times \frac{\Delta L}{L}\right)$
- $\frac{\Delta\rho}{\rho} = \left(\frac{10}{590}\right) + 3\left(\frac{0.1}{6.0}\right)$ [1 mark]
- $\frac{\Delta\rho}{\rho}$ = 0.0669 = 0.07 [1 mark]

Quote the density and uncertainty to an appropriate amount of sig figs:

- Actual uncertainty in ρ: $\Delta\rho$ = 2.73 × 0.07 = 0.18 = 0.2 g cm^{-3}
- Density = 2.7 ± 0.2 g cm^{-3} [1 mark]

[Total: 4 marks]

Remember, the uncertainty must always be quoted to at least 1 sig fig less than the quantity itself - this is because you cannot know the uncertainty in a quantity to a greater precision that the quantity itself!

3d

(d)

(i) The type of error is:

- Systematic; [1 mark]
- (Because) all readings have the same error **OR** this error cannot be eliminated by repeating and averaging; [1 mark]

(ii) The readings are precise but not accurate because:

- Vernier callipers measure to a fraction of a millimeter / resolution of 0.1 mm, so the readings are precise **OR** if repeated, readings will be consistent / almost constant; [1 mark]
- But, all readings will have the same error, hence they are not accurate; [1 mark]

[Total: 4 marks]

4a

(a)

(i) A systematic error that can affect both devices is:

- Incorrect calibration (of the device); [1 mark]

(ii) A systematic error that can only affect the analogue voltmeter is:

- Zero error; [1 mark]

(iii) A random error that can affect both devices is:

- Incorrect interpolation between scale readings / divisions; [1 mark]

(iv) A random error that can only affect the analogue voltmeter is:

- Parallax error / incorrectly reading the scale from different angles; [1 mark]

- Incorrect interpolation between scale readings / divisions; [1 mark]

(iv) A random error that can only affect the analogue voltmeter is:

- Parallax error / incorrectly reading the scale from different angles; [1 mark]

[Total: 4 marks]

4b

(b)

(i) Calculate the power dissipated by the resistor:

- (From Fig. 1.1): voltmeter reading = 5.5 V [1 mark]
- Electrical power: $P = \frac{V^2}{R}$
- $P = \frac{5.5^2}{125}$ = 0.242 W [1 mark]

(ii) Calculate the percentage uncertainty in the calculated power:

- Uncertainty in voltmeter reading = half the smallest division = ± $\frac{1}{2}$ × 0.5 = ± 0.25 V
- % uncertainty in voltmeter reading = $\frac{0.25}{5.5} \times 100\%$ = 4.5% [1 mark]
- % uncertainty in power = (2 × 4.5%) + 3% = 12 % [1 mark]

(iii) The power with its uncertainty is:

- Absolute uncertainty in P = 0.12 × 0.242 = ± 0.029 W [1 mark]
- Power = 0.24 ± 0.03 W (2 s.f.) [1 mark]

[Total: 6 marks]

4c

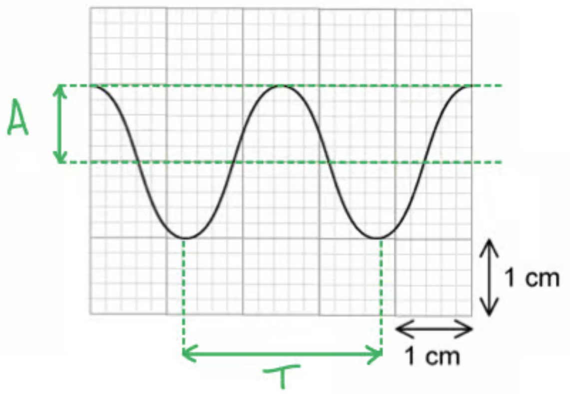
(c)

(i) Use the trace to determine the amplitude:

- Y-plate setting = 2.5 mV cm^{-1}
- Amplitude is measured from the mid-point to the peak of the wave
- (From trace): amplitude reading = 1 large square = 1 cm [1 mark]
- Amplitude, $A = 1 \times 2.5 = 2.5$ mV [1 mark]

(ii) Use the trace to determine the frequency:

- Time-base setting = 0.50 ms cm^{-1}
- Time period is measured from one peak to the next peak
- (From trace): time period reading = 13 small squares = 13 × 0.2 = 2.6 cm [1 mark]
- Time period, $T = 2.6 \times (0.5 \times 10^{-3}) = 0.0013$ s [1 mark]
- Therefore, frequency, $f = \frac{1}{T} = \frac{1}{0.0013} = 769 = 770$ Hz [1 mark]



[Total: 5 marks]

4d

(d)

(i) Determine the % uncertainty in amplitude:

- Y-plate setting = 2.5 mV cm^{-1}
- 1 division = 0.2 cm
- Uncertainty in a reading = half the smallest division = $\pm \frac{1}{2} \times (0.2 \times 2.5) = \pm 0.25$ mV [1 mark]

- % uncertainty in amplitude = $\frac{0.25}{2.5} \times 100\%$ = 10% [1 mark]

(ii) The amplitude, with its uncertainty, is:

- Amplitude = 2.50 ± 0.25 mV [1 mark]

[Total: 3 marks]

5a

(a) Determine the mean period of oscillation and its percentage uncertainty:

- Mean period = $\frac{0.67 + 0.66 + 0.67 + 0.68 + 0.69 + 0.64 + 0.66 + 0.65 + 0.68 + 0.65}{10}$
- Mean period = 0.67 s [1 mark]

Calculate the percentage uncertainty:

- Uncertainty in period = ± half the range = ± $\frac{1}{2}$ (0.69 – 0.64) = 0.025 s [1 mark]
- Percentage uncertainty in T = $\frac{0.025}{0.67} \times 100\%$ = 3.7% [1 mark]
- Therefore, the mean period = 0.67 s ± 3.7 %

[Total: 3 marks]

5b

(b)

List the known quantities:

- Uncertainty in L, ΔL = 1.8%
- Uncertainty in g, Δg = 1.6%

Calculate the time period, T:

- Time period, T = $\frac{total\ time\ of\ oscillations}{number\ of\ oscillations}$

- $T = \frac{18.4}{20}$ = 0.92 s [1 mark]

Calculate the absolute uncertainty in *T*:

- $T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\left(\frac{L}{g}\right)^{\frac{1}{2}}$
- % uncertainty in *T* = $\left(\frac{1}{2} \times 1.8\right) + \left(\frac{1}{2} \times 1.6\right)$ = 1.7% [1 mark]
- Uncertainty in *T* = 0.017 × 0.92 = 0.016 s (2 s.f) [1 mark]
- Therefore, *T* = 0.92 ± 0.016 s

[Total: 3 marks]

Break this question down into stages and make sure you propagate the uncertainty correctly.

When propagating the uncertainty containing a square root of a quantity, remember that the square root of a number is the same as putting it to the power of ½.

5c

(c) The readings are subject to systematic uncertainties because:

- The graph has a y-intercept; [1 mark]

The readings are subject to random uncertainties because:

- Points are scattered around the line of best fit; [1 mark]

[Total: 2 marks]

5d

(d) Determine the fractional error in *k*:

List the known quantities:

- Fractional error in *T*, $\frac{\Delta T}{T} = \alpha$

- Fractional error in m, $\frac{\Delta m}{m} = \beta$

[Total: 2 marks]

5d

(d) Determine the fractional error in *k*:

List the known quantities:

- Fractional error in *T*, $\frac{\Delta T}{T} = \alpha$
- Fractional error in *m*, $\frac{\Delta m}{m} = \beta$

Write an expression for the uncertainty of *k* in terms of *α* and *β*:

- Period of a mass-spring: $T = 2\pi\sqrt{\frac{m}{k}} \quad \Rightarrow \quad k = 4\pi^2\frac{m}{T^2}$ [1 mark]
- $\frac{\Delta k}{k} = 2\frac{\Delta T}{T} + \frac{\Delta m}{m}$
- $\frac{\Delta k}{k} = 2\alpha + \beta$ [1 mark]

[Total: 2 marks]

Remember the rules for propagating errors. These must be applied carefully to the rearranged form of this equation. These are:

- Values that are added or subtracted:
  - Add their **absolute** uncertainties
  - In this example, there are no quantities that are added or subtracted
- Values that are multiplied or divided:
  - Add their **percentage** or **fractional** uncertainties
  - In this example, *m* and *T* are divided so we can add %Δ*T* + %Δ*m*, or the fractional uncertainties (as shown in the mark scheme)
- Values that are to a power:
  - Multiply their **percentage** or **fractional** uncertainty by the power

- In this example, T is squared, so its uncertainty is $2\Delta T$